

Laptop Sales and Purchase system

(SRS)

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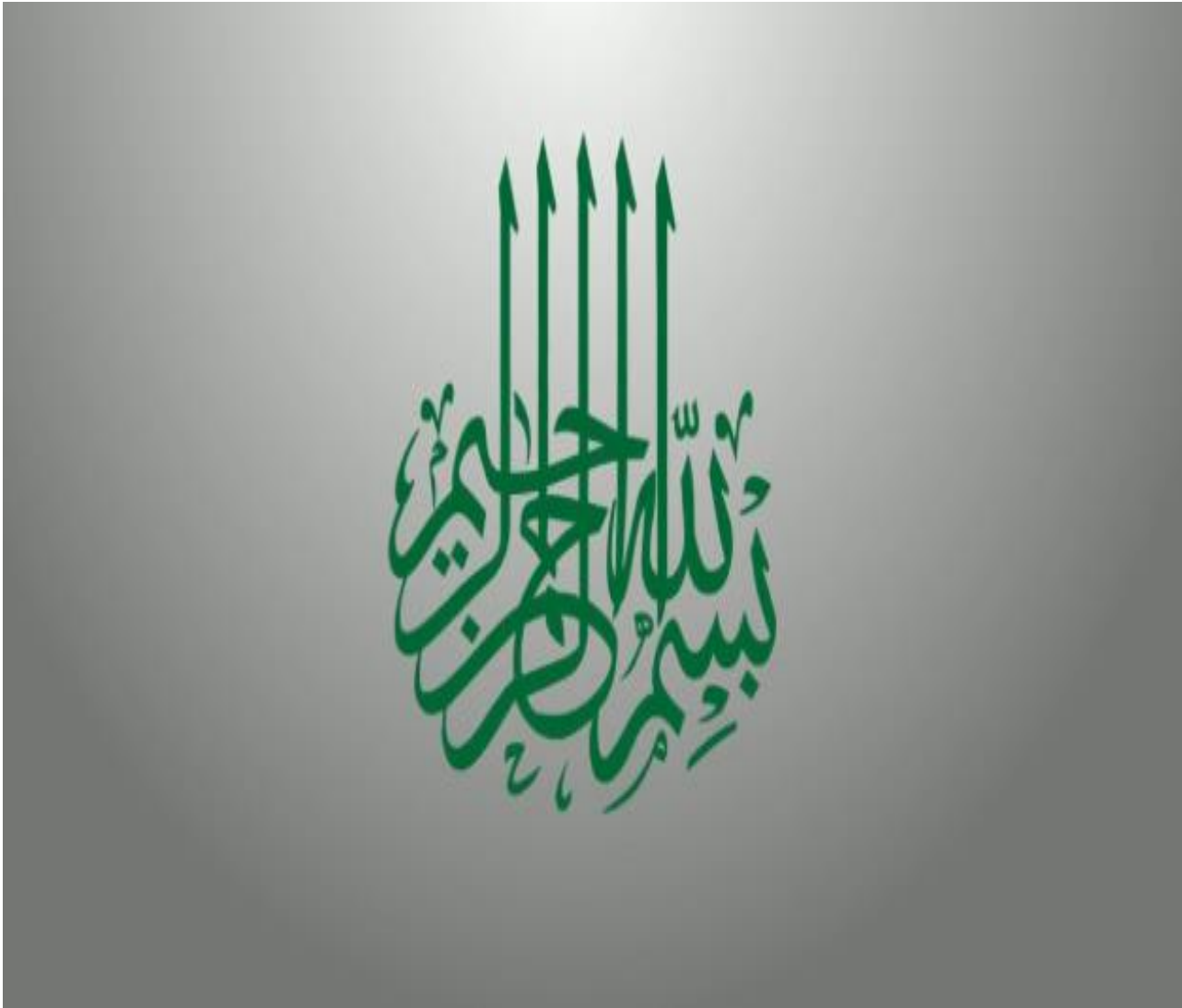
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NATIONAL COLLEGE HASILPUR





[Laptop Sales and purchase System]

SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

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CHAPTER NO 1:

INTRODUCTION

1. Introduction

“**Laptop Sales and Purchase system**” is used to keep record of whatever the sale from shop. It is also used keep track of remaining balance. It is the project of selling and purchasing the laptop by using this software the shop manager can easily manage their whole system. It provide the faculty of data update, data recovery, data manipulation, data modification and integration. It also provide data security from unauthorized access. “**Laptop Sales and Purchase system**” also saves times like in traditional data system. Data recovery and backup is also a major benefit of this system in case of any loss of data. This SRS is going to define the different functions, interfaces and performance of the system. This document is a Desktop- based that is used for the Laptop management system

1.1 Purpose

Main purpose of laptop management system is to better performance of system so that every person an easily interact and perform their actions. Information about the Laptops and their records are easily kept in this system. Laptops are fully functional that consume less power. Actually this system is only made to access all the information. All records are permanently stored in it .Time saving and data correctness is the main benefit of this system. We can have the record of sale and purchase system, so we can deal the customers easily. The main purpose of sale and purchase is the time and place where a retail transaction is completed. We usually preferred our laptops for gaming purposes.

Stock lists and its copy

- Manage costs (internal and external)
- Sales of Laptop on Finance fully supported
- A high quality sales printed
- Unlimited laptop photo graph
- Complete profit and loss analysis on every laptop sold

1.2 Scope

In this subsection:

- Best Seller(The bestseller can sell the laptops in best price)

- Bundling of products(The customer can purchased the big box electronics with it's economic of scope)
- Social media marketing(Through laptops we can do the social media marketing)
- Strategy for information (We can have the purchase price and which also requires monthly payment through a subscription)

1.3 Definitions, Acronyms, and Abbreviations.

- **WWW** (worldwide web and Set of large interlinked document on internet).
- **SQL** (Structure Query Language).
- **GUI** (Graphical User Interface).
- **Admin** (Login for attraction with application).
- **Modify** (For modification in database and user records).
- **Logout** (For un-attraction with application)
- **RESUME** (To restart something).

1.4 References

In this subsection:

- www.google.com.
- www.w3school.com.
- www.draz.pk.
- www.wikipedia.com

1.5 Overview

The rest of the document will explain the specification of laptop management system in detail. SECTION 2 will explain the general factors that affect the software and its requirements such as project constraints and user characteristics etc. SECTION 3 will explain the detailed functional specific system and other related requirements of the system.

CHAPTER NO 2:

THE OVERALL DESCRIPTION

2. The Overall Description

2.1 Product Perspective

List each system interface and identify the functionality of the software to accomplish the system requirement and the interface description to match the system. These are external systems that you have to interact with. For instance, if you are building a business application that interfaces with the existing employee payroll system what is the API to that system that designer's will need to use.

2.1.1 USER INTERFACES

It may consist of following:

- The logical characteristics of each interface between the software products and its users.
- All the aspects of optimizing the interface with the person who must use the system.

This is a description of how the system will interact with its users. Is there a GUI, a command line or some other type of interface? If you are designing for general student population for instance, what is the impact of ADA (American with Disabilities Act) on your interface?

2.1.2 Apportioning of Requirements

Identify requirements that may be delayed until future versions of the system. After you look at the project plan and hours available. You may realize that you just cannot get everything done. This section divide the requirements into different sections for development and delivery.

2.2 Product Functions

The software product will perform the following functionality:

- Customer gives information to shop manager.
- Customer pays all the bills.
- Shop manager gets customer requirements.
- Manager manipulate all gathered information.
- Manager displays laptop bills.

- Manager stores all information of customer.

2.3 User Characteristics

The user of this document should have basic knowledge of searching over the web browser should have the knowledge that how to view history, how to book mark web pages, how to use different tabs, how to download the material. User should also have knowledge of word editing so that he could use these feature in detail. So we can distribute the use in two levels:

- **EDUCATION LEVEL:** The user should be at least educated to understand the language of browser and editor (English).
- **TECHNICAL LEVEL:** The user should be comfortable using general purpose application on the personal computer.

2.4 General Constraints

Software has the following constraints that should be kept in mind by developer while developing the software:

- Font color and sizes need to be changeable to meet the requirements of sight damaged viewers.
- The situation become more complex when collaborative resizing of windows and frames, printing or communication condition.
- Internet speed is major issue in the ensuring the good quality of web browsing and downloading.

2.5 Assumptions and Dependencies

Some web pages display wrong with some browsers:

- Browser specific codes are used in programing of web masters.
- Processing power of CPU also effect the performance of system
- The user must know the English Language.

2.5.1 COMMUNICATION INTERFACES

Specify the various interfaces to communicate such a local network protocols. If you are using a custom protocol to communicate between systems then document that protocol here to designers know what to design. If it is a standard protocol you can reference an existing document or RFC.

2.5.2 MEMORY CONSTRAINTS

Specify any applicable characteristics and limits on primary and secondary memory. If all the customers' machine have only **128K** of RAM, then your target design has got to come in under **128k** so there is an actual requirement. "Focus groups have determined that our target market has between **256-512M** of RAM therefore the design footprint should not exceed **256M**". If there are no memory constraints.

2.5.3 OPERATIONS

Specify the normal and special operations required by the user such as:

- The various modes of operations in the user organization.
- Data processing support functions
- Periods of interactive operations and periods of unattended operations.
- Backup and recovery operations.

2.5.4 PRODUCT FUNCTIONS

Provide a summary of major functions that the software will perform. This describes the functionality of the system in the language of the customer. What specifically does the system that will be designed have to do? Drawings are good but remember this is a description what the system needs to do, not how you are going to build it.

2.6 CONSTRAINTS

Provide a general description of any other items that will limit the developer's options. These can include:

- Regulatory policies
- Hardware limitations(for example, signal timing requirements)
- Parallel operation
- Interface to other applications.
- Audit functions
- Control functions
- Reliability requirements

- Safety and security consideration

CHAPTER NO 3:

SPECIFIC REQUIREMENTS

3. Specific Requirements

This section contains all the software requirements. These requirements should include at minimum a description of every input into the system, every output from the system and all the functions performed by the system in response to an input or in support of an output.

- Specific requirements should be stated with all the characteristics of good SRS
 - Correct
 - Complete
 - Consistent
 - Verifiable
 - Modifiable
- Specific requirements should be cross referenced to earlier documents that relate.
- All requirements should be uniquely identifiable.

3.1 External Interface Requirements

3.1.1 System Interfaces

- A parent page consist on these functions like (login, add stock, modify stock).
- First of all admin must login to interact with system so, login box appear.
- Admin select category of laptop like (Dell, Lenovo and HP etc.).
- When admin select category then appear a box with appropriate models of Laptop.
- Then select a laptop and generate the bill of this laptop.
- At the end system must logout after use.
- When manager will click on stock button then all stock of laptop from database will show on one form.

- When you will click on sample image icon from menu bar images of all laptop Models with prices will show.
- When user will click on logout all forms will close and just login form will show on the screen.
- When manager give username and password and click on login the main view will open and login form will un –visible.

3.1.2 Interfaces

- As you know that main page will consist on some functions like login.
- When seller signup its data it will store in database table, when login we retrieve that data from database
- When seller click on login a form it will open and seller put his email and password and click on login button ,this given data will be check in the database
- For signup purposes email password must required
- If signup is successful seller can submit his adds request to admin with login
- Admin can see all requests that will be done by username and password

3.1.3 Hardware Interfaces

- A system must contain 1GB RAM is required.
- 2.16 GHz processor is required.
- 233 MB Hard Disk Drive will be used.

3.1.4 Software Interfaces

- SQL as an database and OS (Window)
- Visual studio on OS (Window)

3.1.4.1 Microsoft SQL Server

Our website must use SQL Server as its database component. Manager can setup and modify the database.

3.1.5 Communications Interfaces

- The protocol that is used for communication is SQL.
- Communication interfaces required by this local SQL server.
- It must specify any security issues, Data transfer rates and synchronization mechanisms.

3.2 Functional Requirements

A Functional requirement defines a function of a system or its components, where a function is described as a specification of behavior between outputs and inputs. Our system provide these facilities;

3.2.1 Admin Login

3.2.2.1 Introduction

In this admin will login for access the system

3.2.2.2 Inputs

Email

Password

3.2.2.3 Processing

System will check the email and password from database

3.2.2.4 Output

Main view will open

3.2.2.5 Error Handling

Wrong email or password

3.2.2 Laptop Records

3.2.2.1 Introduction

In this step stock will be store in database

3.2.2.2 Inputs

Price of laptops

Stock of laptops

Model of laptops

3.2.2.3 Processing

Check the valid entry values

3.2.2.4 Outputs

Message will be shown that record added

3.2.2.5 Error Handling

Something missing like price or model

3.2.3 Stock Updated

3.2.3.1 Introduction

In this section stock will be updated

3.2.3.2 Inputs

Id

New stock

3.2.3.3 Processing

System will check is this id exist or not

3.2.3.4 Output

Message will be shown stock updated

3.2.3.5 Error Handling

Such id no exist

3.3 Use Cases

This section contains use case of the “Laptop Sales and Purchase System”

3.3.1 Use Case#1 Admin Login

This part for use case diagrams for “LAPTOP SALES AND PURCHASE SYSTEM”.

Description:

Admin must be login to access this system.

Actor:

Admin

Precondition:

Admin must be registered.

Basic flow:

- User enter his/her username and password.
- System check this data with database.
- If username and password is correct.
Then user access our system
Else
User can't access our system.

Alternative flow:

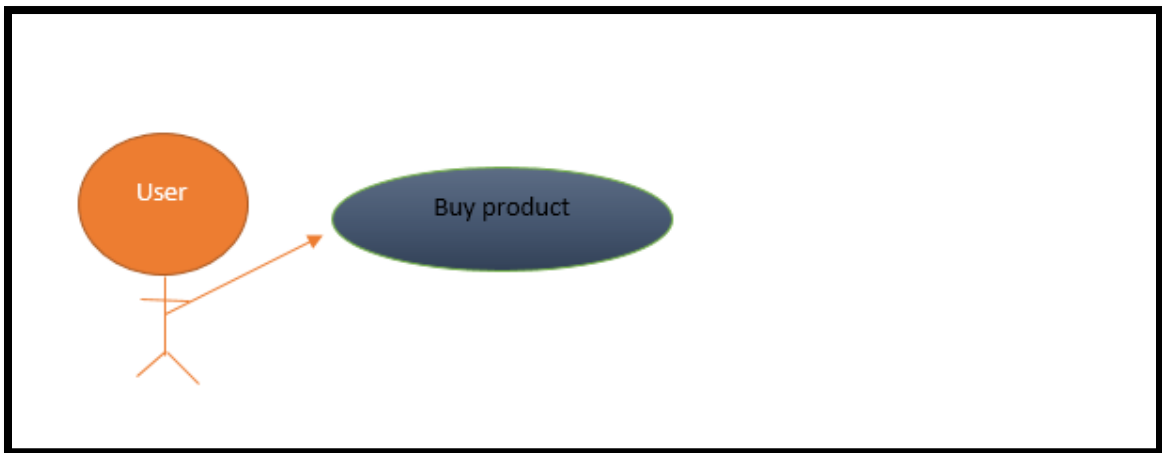
There is no alternate flow.

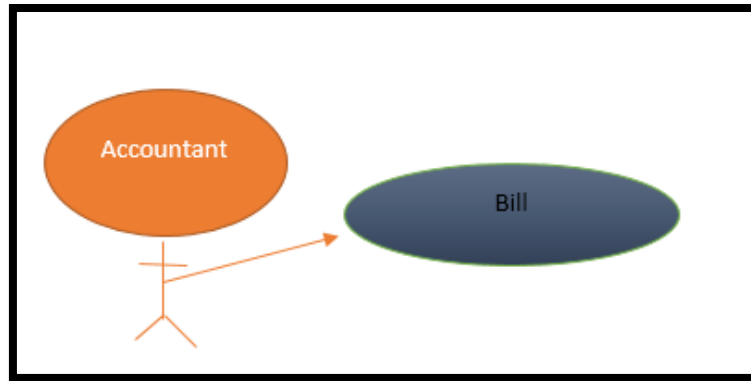
Post condition:

User should be login.

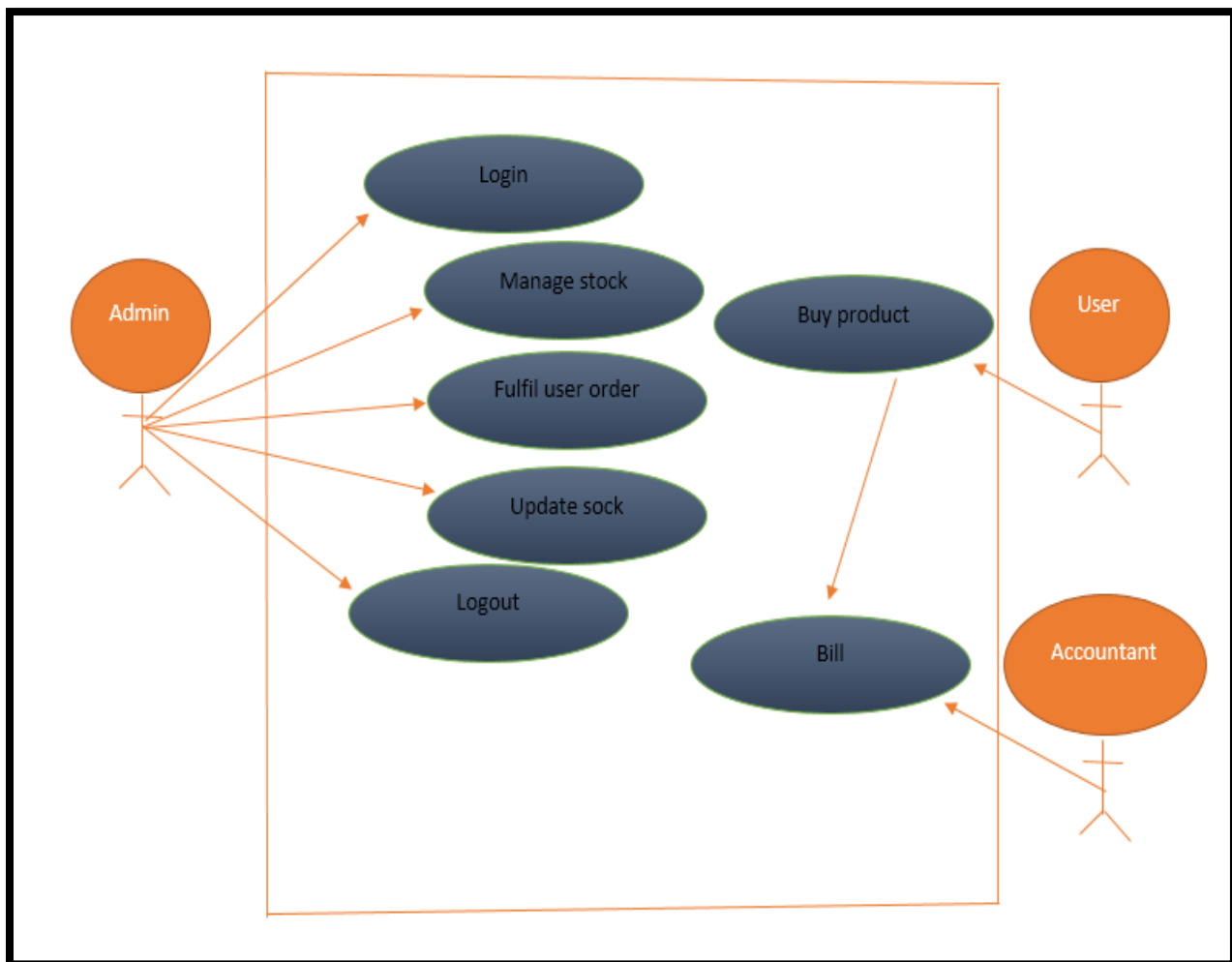
Exception:

Username or password is wrong





3.3.2 Use case# 2Add Product



Description:

Admin add the products.

Actor:

- Admin

Precondition:

Admin must be login.

Basic flow:

- Admin enter some specific data like mobile name, picture, model and price etc.
- System check that data is valid or not. Or fields can't be blanked.
- If data is valid then store it in database
Else
- Your data does not store. Or your field cannot be filled properly.

Alternative flow:

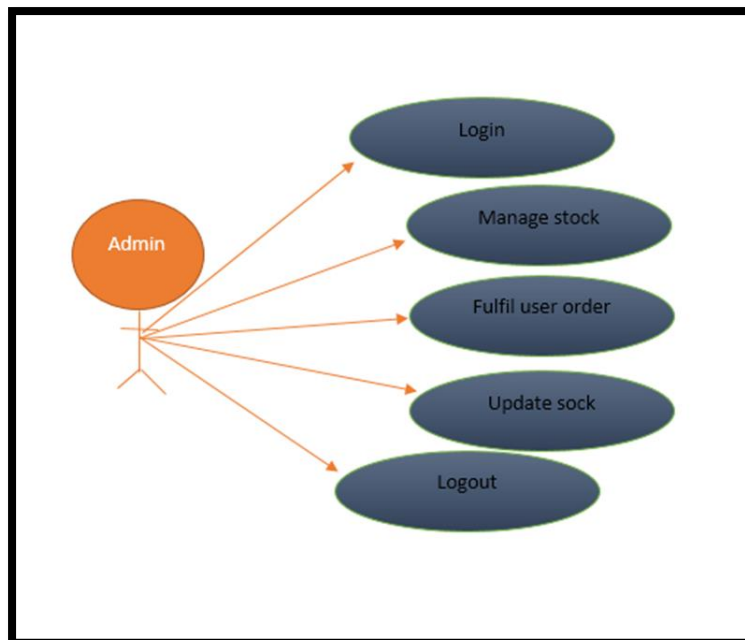
There is no alternate flow.

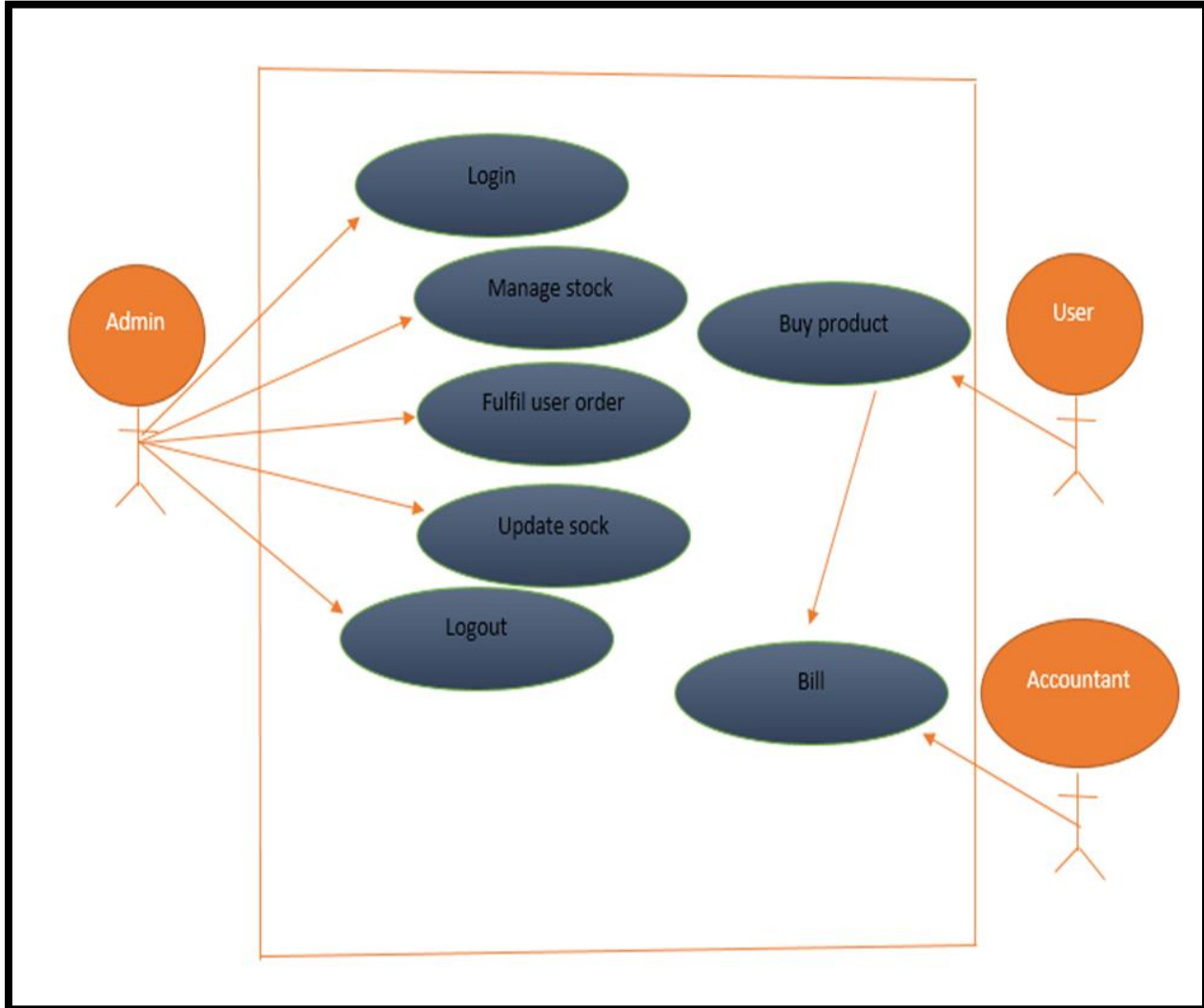
Post condition:

User see products on shop or main form after login process.

Exception:

product does not stored.





3.3.3 Use Case#3 Remove Customer Records or product

Description:

Admin remove the customer record from database or remove the products.

Actor:

- Admin is actor to perform functionality.

Precondition:

Admin should login.

Basic flow:

- Admin enter the id of product or customer.

- If id exist in database then appear record of against id.
 - Then admin can remove it
- Else

This id does not exist in database.

Alternative flow:

There is no alternate flow.

Post condition:

- Category or product must be removed.
- This id does not exist in database.

Exception:

Id does not verified.

Manager login	with username and password
Modify customer record	Manager can modify customer record and also can delete it
Modify inventory record	Manager can update stock
Check daily sale	As customer info is stored with date we can also check sale
Buy product	Customer can buy laptops
Generate bill	Manager will put all customer data in form and system will generate bill

Customer detail	Every customer detail will store in database table
-----------------	--

3.4 Classes / Objects

3.4.1 <Admin>

3.4.1.1 Username, Password.

3.4.1.2 Admin login is for interaction with system

3.4.2 <Buyer >

3.4.2.1 Name, CNIC, Address, City, Cell Number.

3.4.2.2 for storing information of buyers

3.4.3 <All laptop company tables >

3.4.3.1 Laptop stock

3.4.3.2 for storing laptop & stock

3.5 Non-Functional Requirements

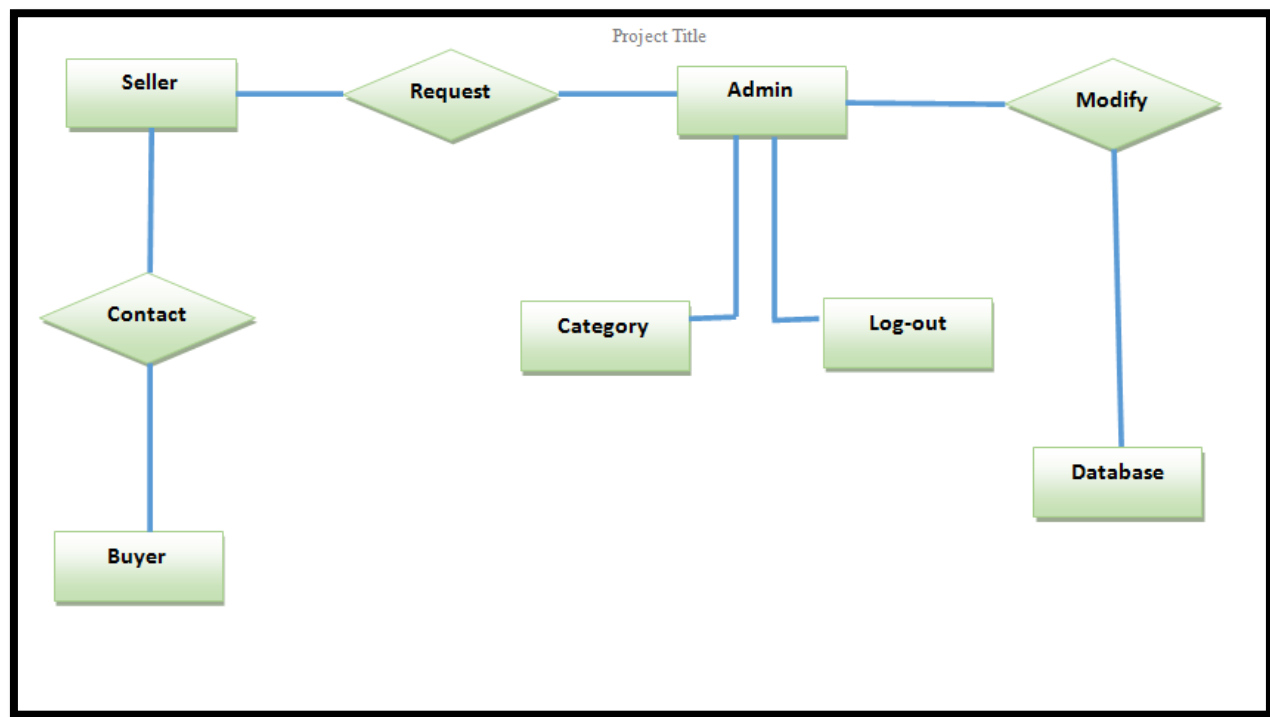
- Interface well design.
- Application is easy to understand.
- Application is easy to handle.
- System must have recoverability skills.
- Performance is very fast.
- Doesn't store incorrect data.
- Data loss can't acceptable.
- System must be reliable.
- System must secure from unauthorized person.
- System must secure from risk.
- System must have maintainability.

3.6 Inverse Requirements

- CNIC consist on 13 length in it
- Laptop model just can be selected from our given models
- Email must be in this format Username@hotmail.com , Username@yahoo.com.

3.7 Logical Database Requirements

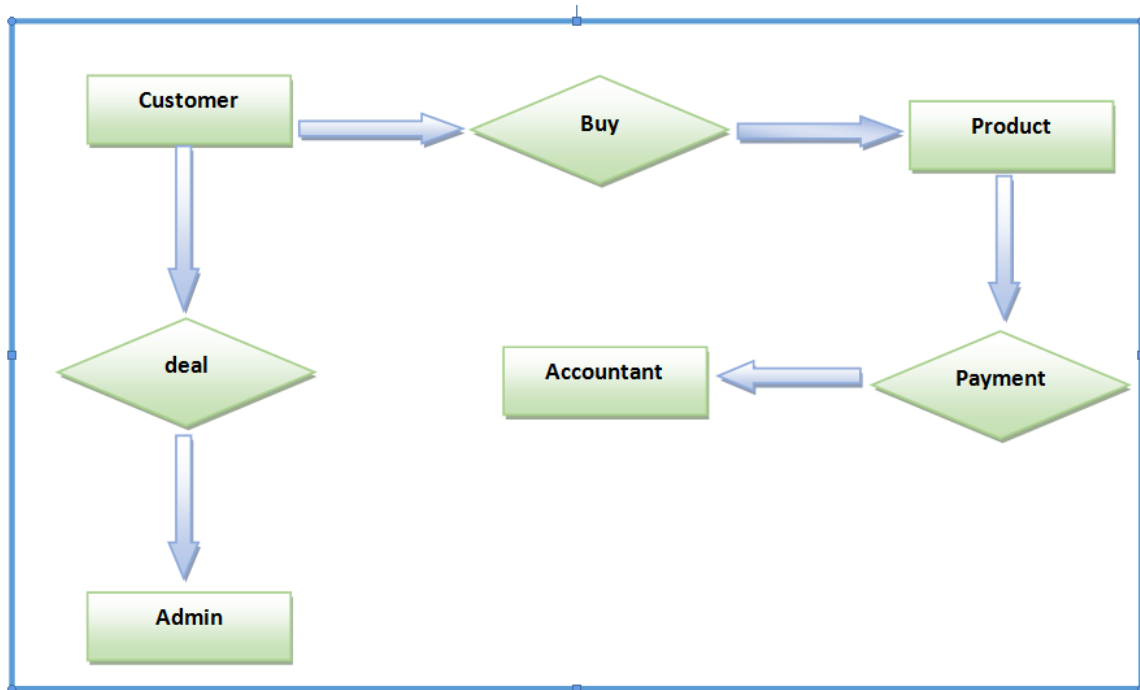
- Database will valid for this project
- Our database keep all record of our system.
- User can't interfere in database.
- There will be store 3tables in database admin authentication, Customer record's product's record.



3.8 Design Constraints

3.8.1 Standards Compliance

- All fields will be filled correctly
- If CNIC length is less than or greater than 13 data will never store but error message show
- Customer data will store in database.
- After completing any operation display a message of successful or not shown.
- All necessary fields must be filled.

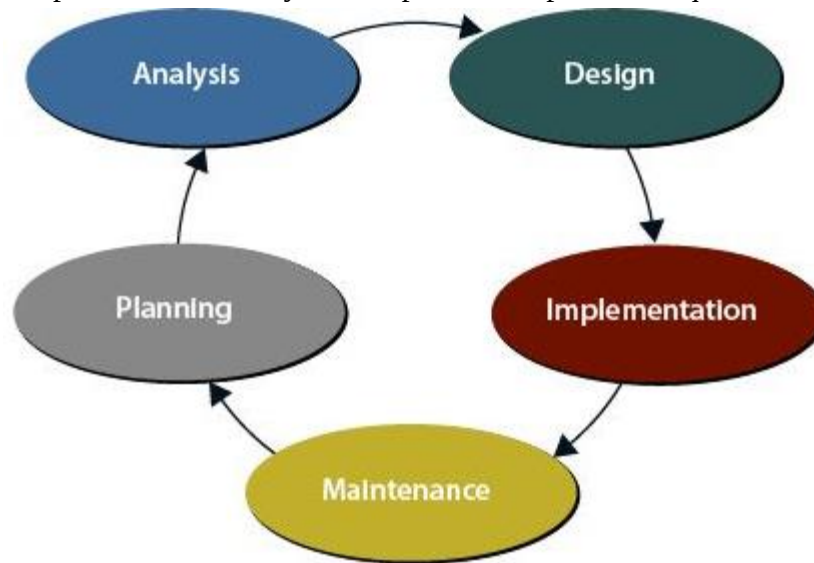


CHAPTER NO 4:

ANALYSIS MODELS

4. Analysis Models

Model based analysis is a method of analysis that uses modeling to perform the analysis and capture the results. The process used the system improvement process, requires model based analysis to



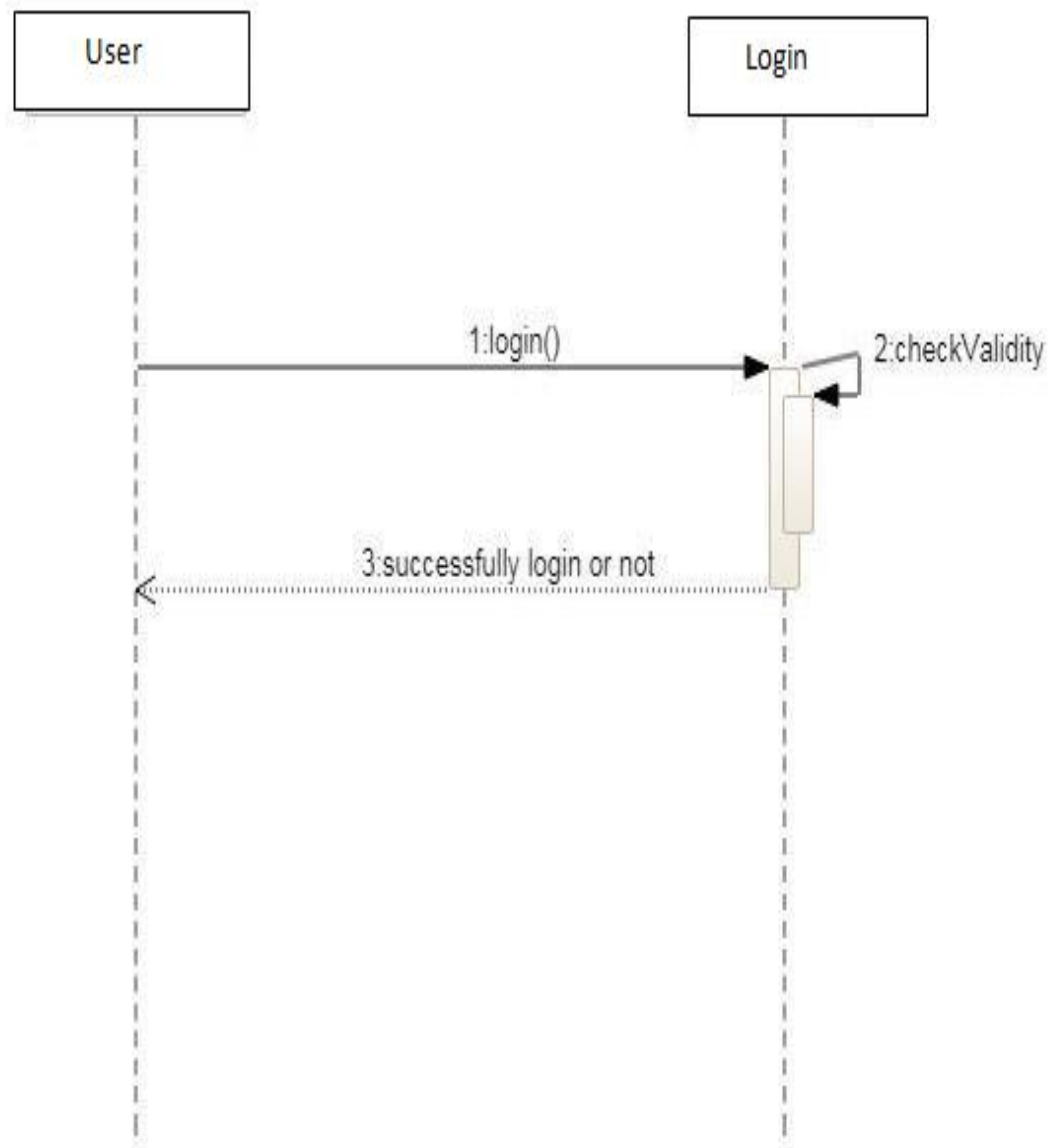
execute the process.

4.1 Sequence Diagrams

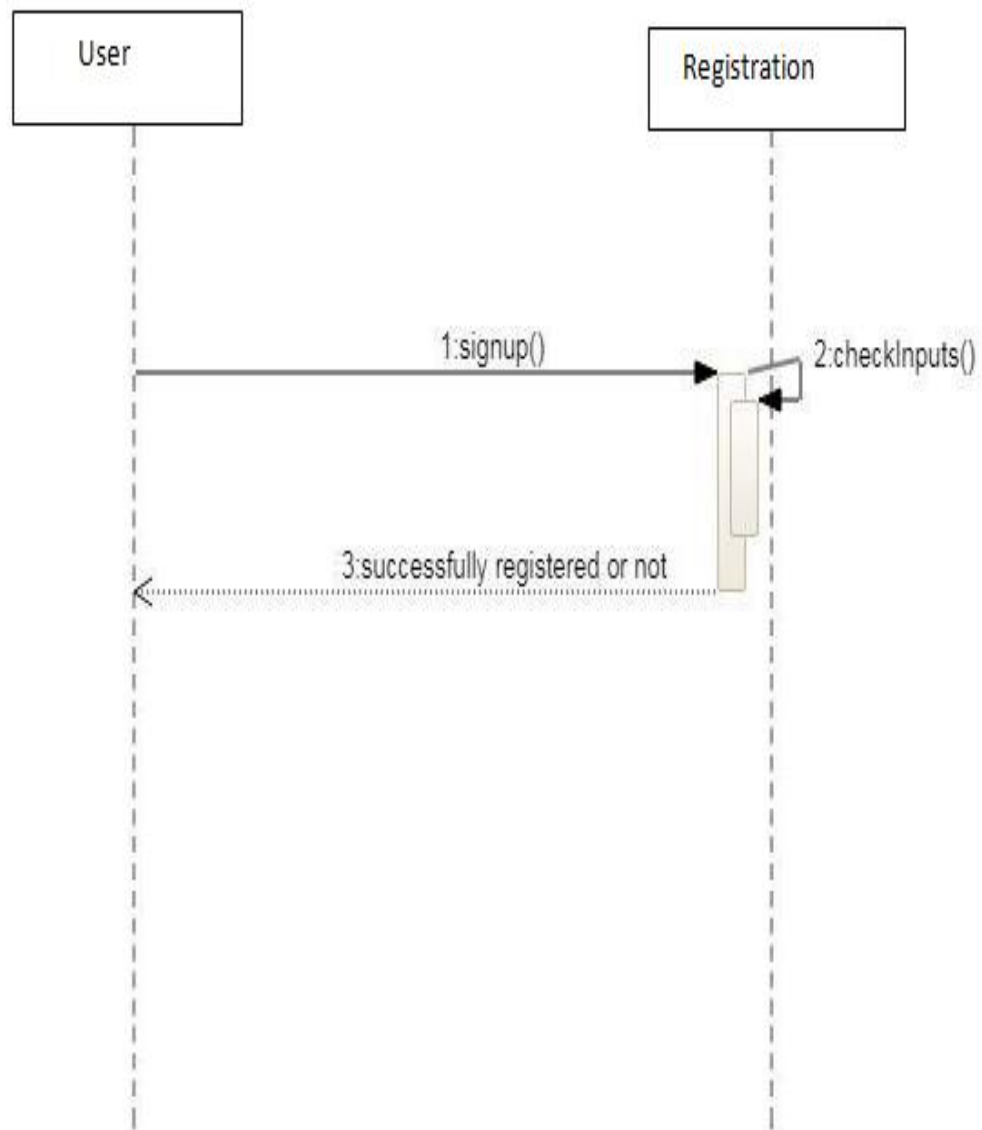
Sequence Diagrams are interaction diagrams that detail how operations are carried out

- Interaction diagrams model important runtime interactions between the parts that make up the system.
- A sequence diagram is made up of a collection of participants.
- Participants – the system parts that interact each other during the sequence.
- Classes or Objects – each class in the interaction is represented by its named icon along the top of the diagram.

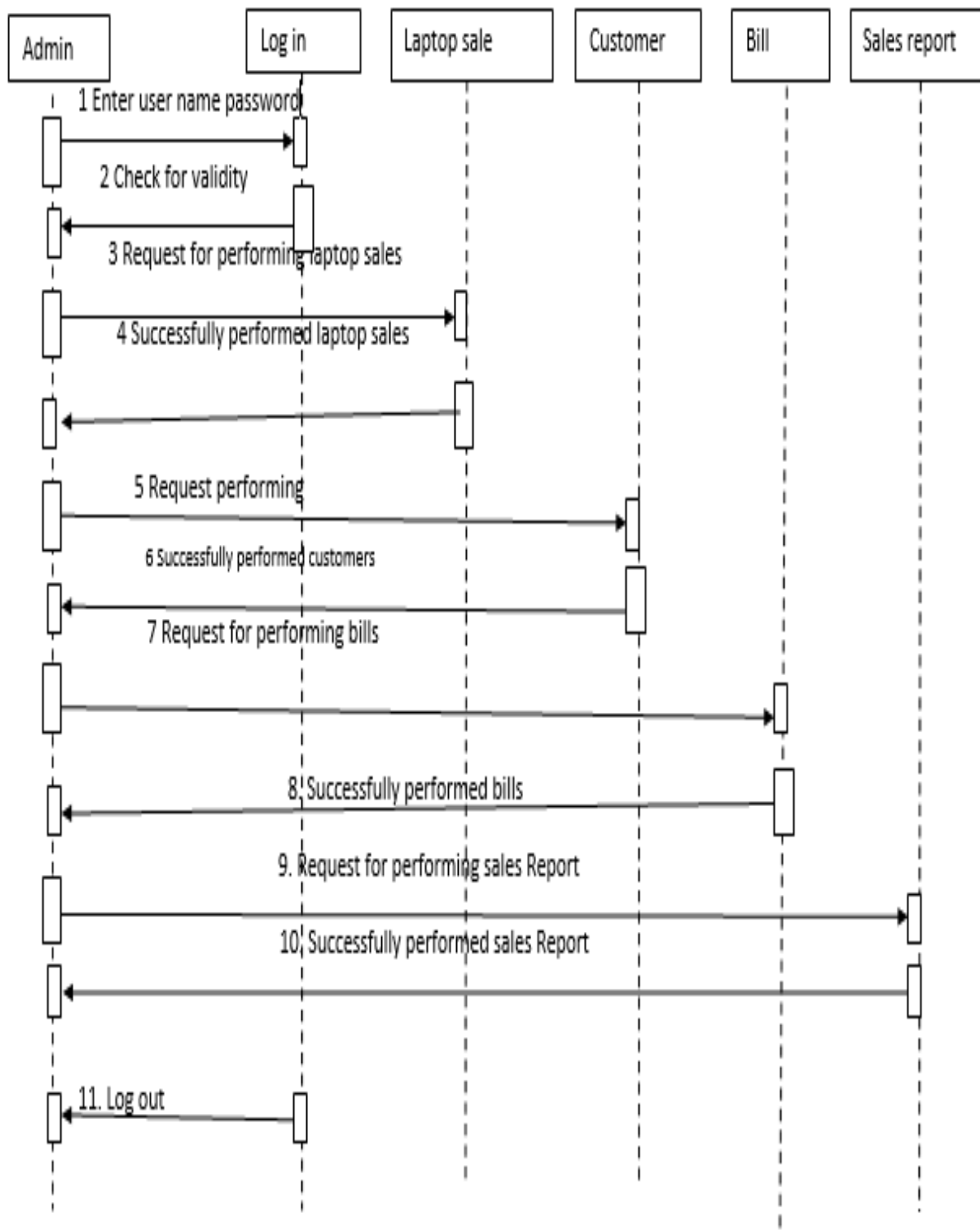
- **Login**

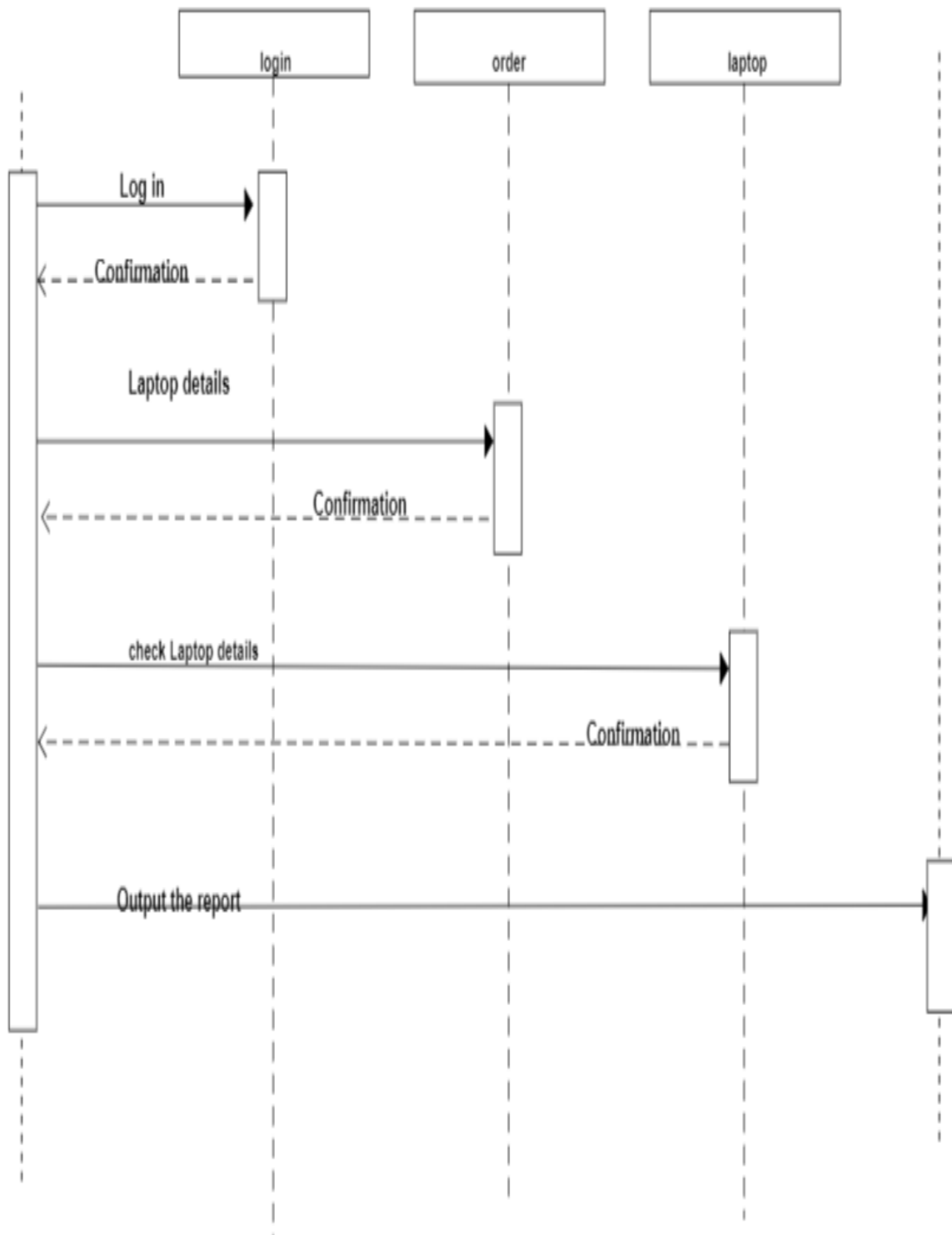


- **Signup**



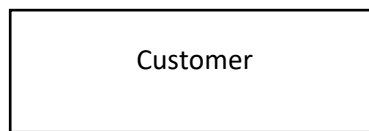
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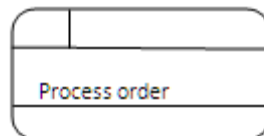


4.2 Data Flow Diagrams (DFD)

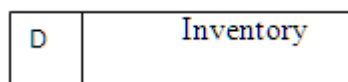
- In a DFD the arrows are actually data transfer between the elements, which are also a part of a system. The primary tools of structured analysis is the Data Flow Diagram (DFD).
- DFD is visible part of diagrams of data flow model technique in the entire system.
- The DFD is a diagram that consists of information that flows through the system using several diagrams.
- So the four symbols of the external entity the data flow, process, and the data store.
- A physical flow can be shown on the input, process and output of the DFD of current system.
- A data flow diagram has no control flow, there is no decision rules and no loops in it.
- **Symbols:**
- **External Entity:**



- **Process**



- **Data store:**



- **Data flow:**

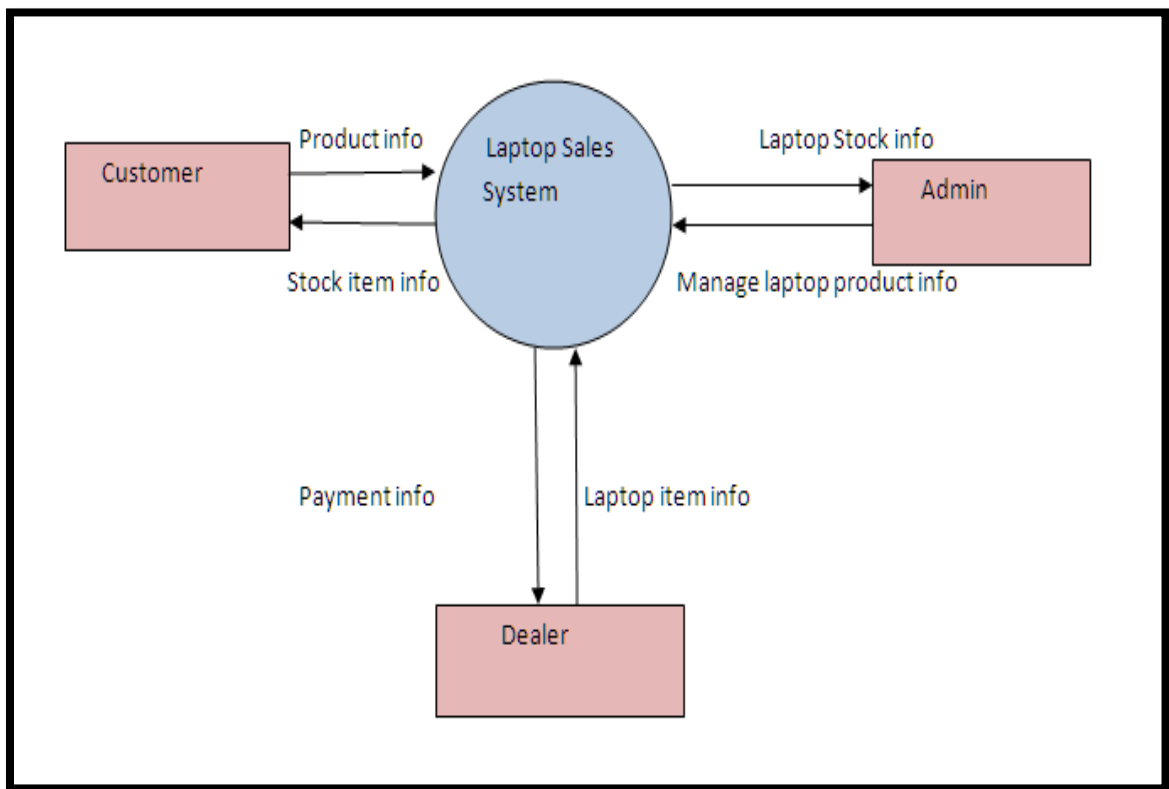


It may consist of follows:

- Context level
- 0- level Diagram
- Detail Level Diagram

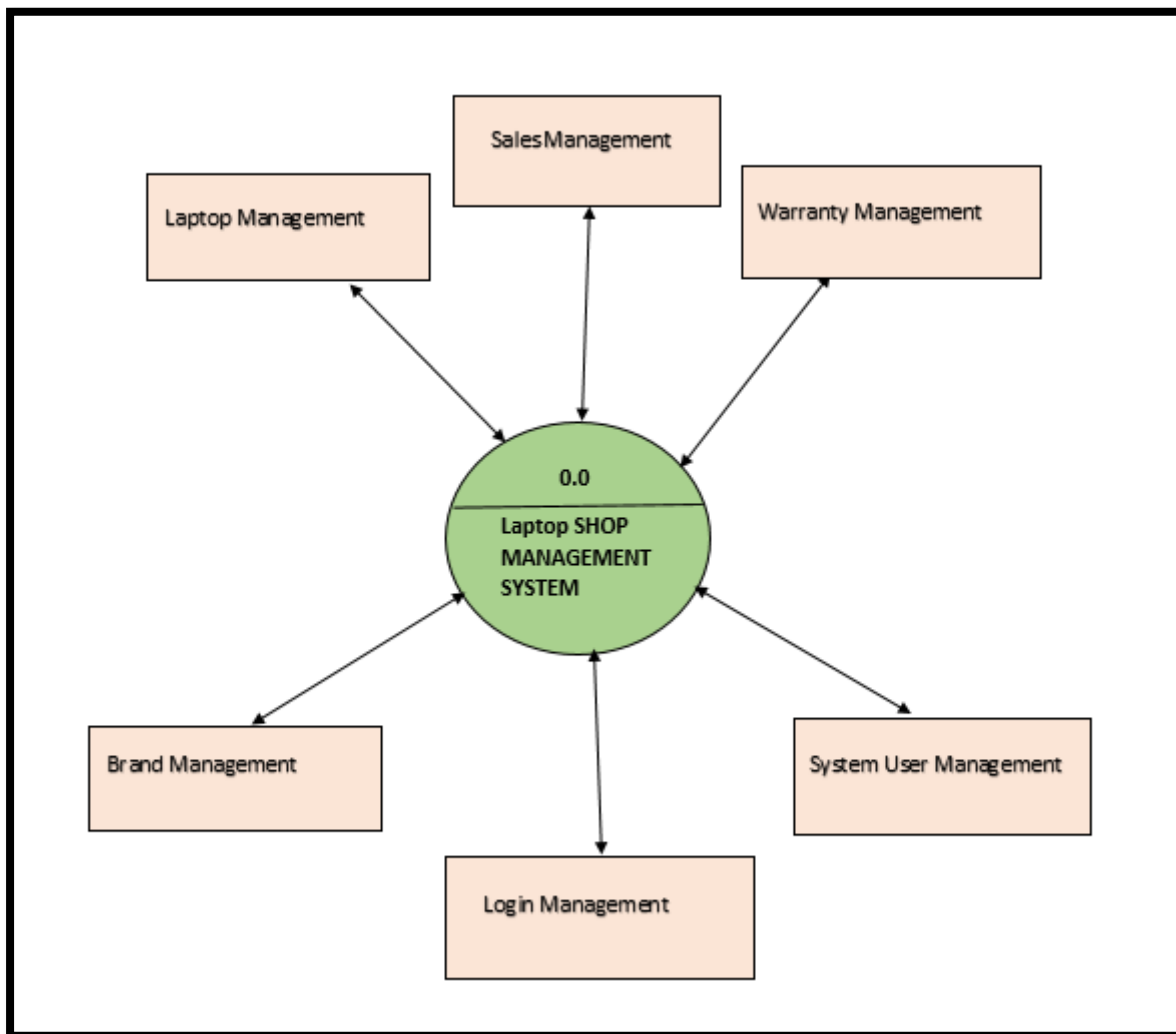
4.2.1 Context level

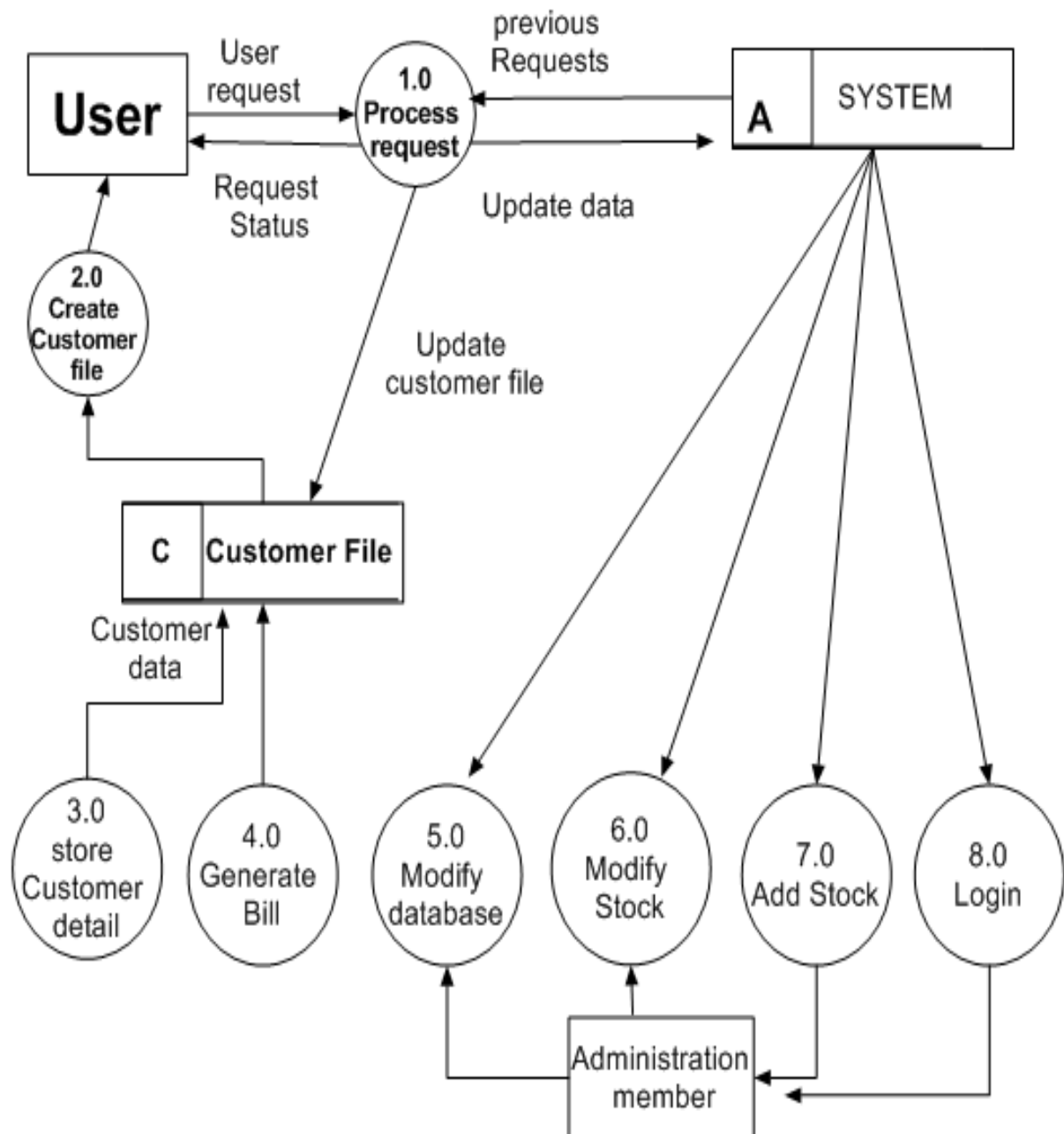
- Context level is a top level data flow diagram. It only contains one process node that generalizes the function of the entire system.
- A context level diagram is drawn in order to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.in external entities.
- It defines the boundary between the system and the part of the system.
- It exchanges data with outside entities. They are often used for high level planning in it.



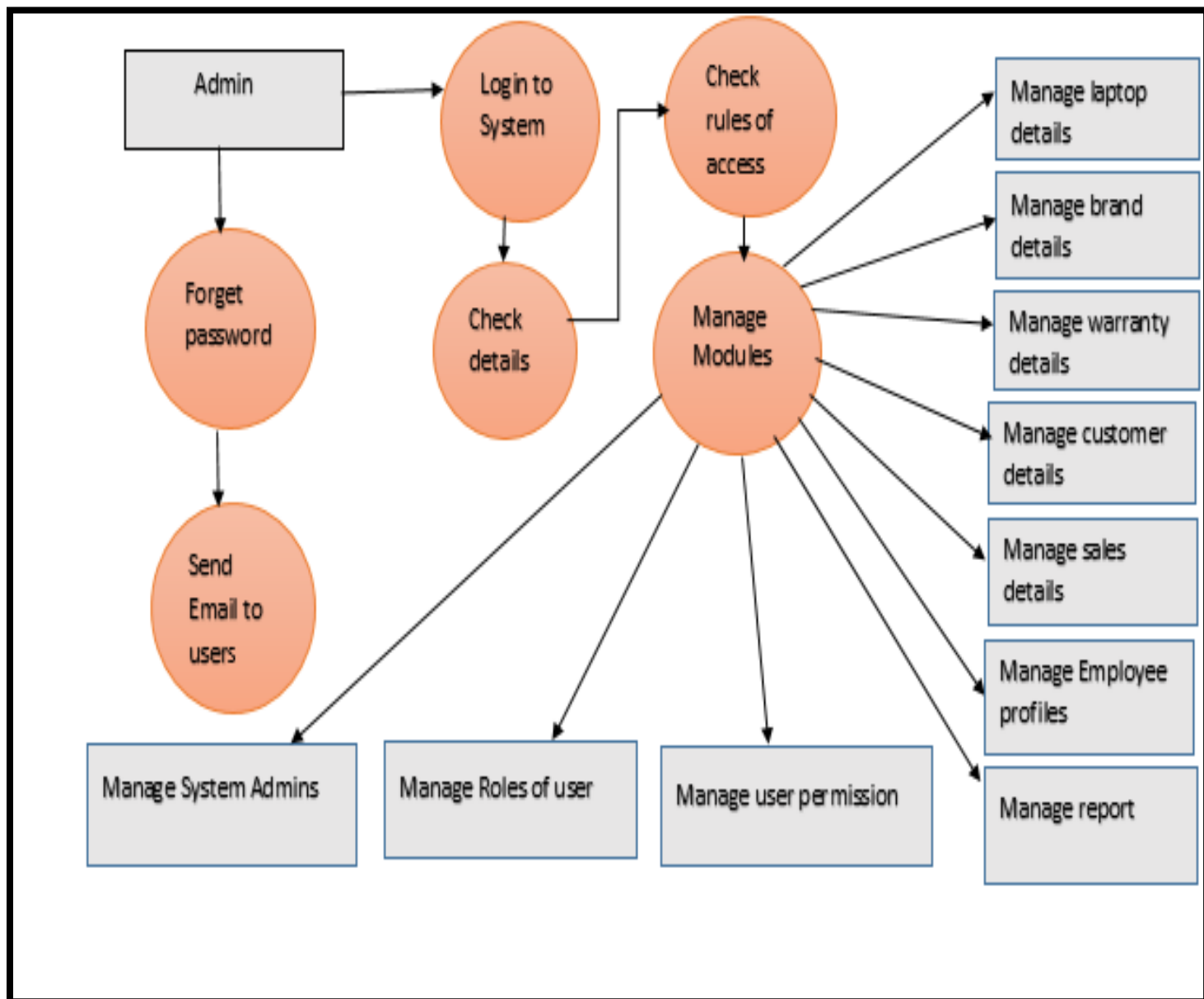
4.2.2 0- level diagram

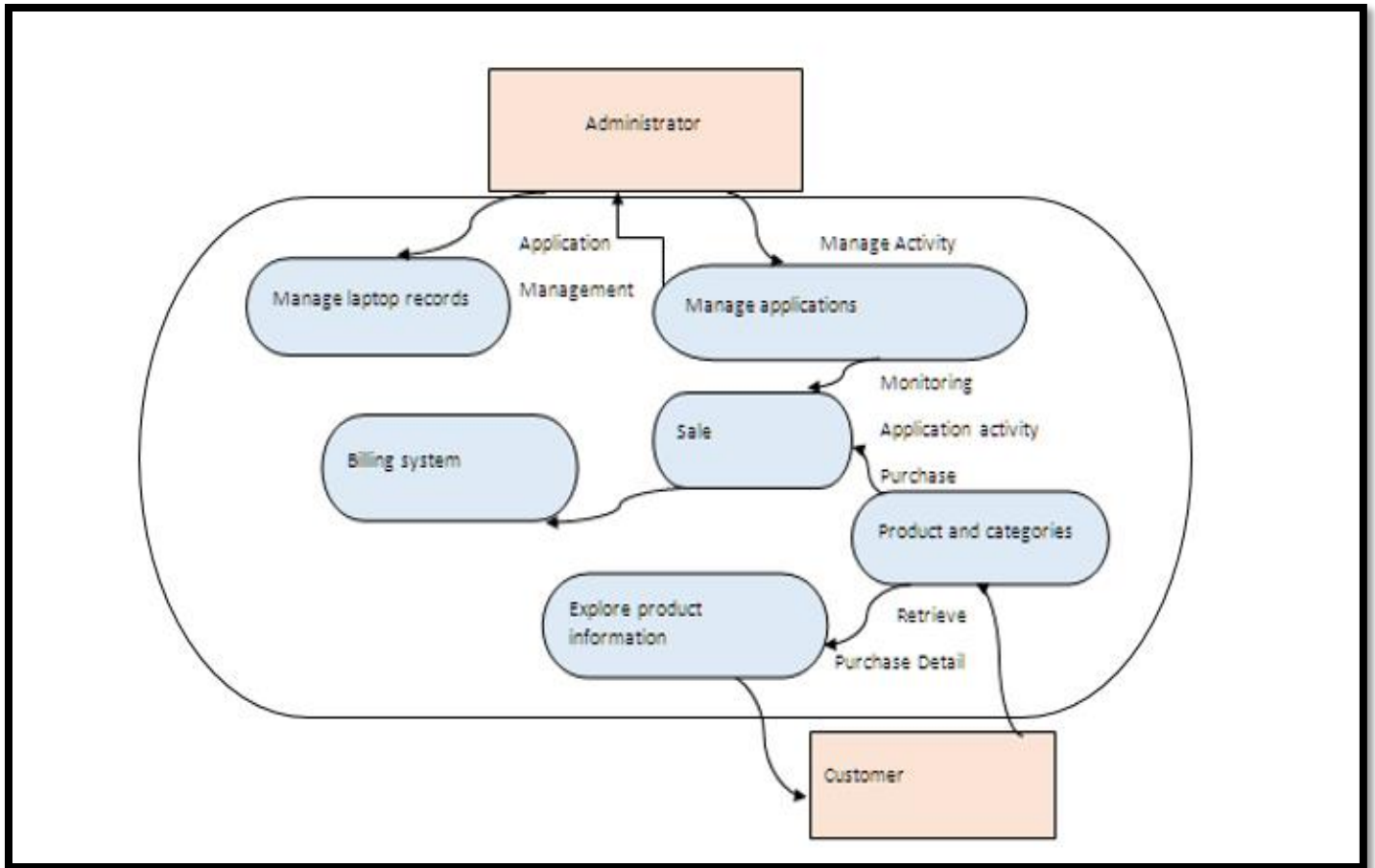
- (0-level) data flow diagram may only contains one process node (process-0) that generalizes the function of the entire system in relationship to external entities.
- A (0- level) diagram is drawn in order to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.
- Level-0 context diagrams shows a system as a single process with inputs and outputs flowing to or from external entities.





4.2.3 Detail level Diagram





4.3 Activity Diagrams

Activity diagram is similar to a flow chart. Activity diagram and state chart diagrams are related. While a state chart diagram focuses on an object undergoing a process (or on a process as an object), an activity diagram focuses on the flow of activities involved in a single process. The activity diagram shows how these single-process activities depend on one another. Activity diagrams can be divided into object swim lanes that determine which object is responsible for an activity.

Introduction of Activity Diagrams:

- Activity diagrams are typically used for business process modeling, In this modeling the logic captured by a single use case or usage diagram, for modeling the detailed logic of a business rule.
- Although, UML activity diagrams could model the internal logic of the entire complex operation.

- It would be better to simply rewrite the operation so that it is simple enough that you don't require an activity diagram in many ways in it.
- UML activity diagrams are the object-oriented equivalent of flow charts and data flow diagrams from structured development of the system.
- UML activity diagrams refer to software engineering method that is used to describe work flows, business processes and other procedures.
- **Activity Diagram**

UML activity diagrams are a kind of behavior diagram, i.e. is a kind of UML behavioral modeling. UML activity diagrams are somewhat similar UML state machine diagrams.

Let's start by describing the basic notations.

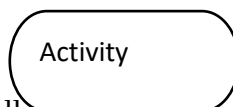
- **Initial node:** The small filled circle is the starting point of the diagram. An initial node isn't required. Make sure the start point is placed in the top left corner. So, it does make it significantly easier to read the diagram.



- **Activity final node:** The filled circle with a border is the ending point of the diagram. An activity diagram can have zero or more activity final nodes in it.



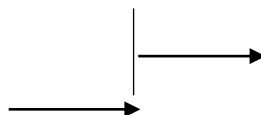
- **Activity:** The rounded rectangles represent activities that occur. You can draw an action state in smart draw using a rectangle with rounded corners.



- **Flow/edge:** They are usually drawn with arrowed line. Action flows are also called edges and paths. So it can do the transitions from one action state to another.

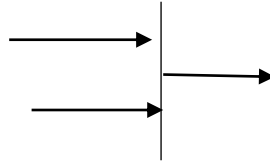


- **Fork:** A black bar with one flow going into it and several leaving it is called fork. This denotes the beginning of concurrent and parallel activity.

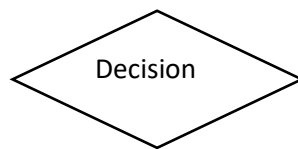




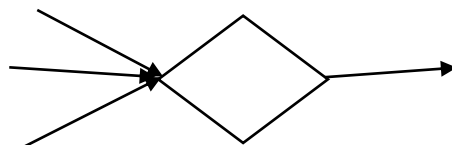
- **Join:** A black bar with several flows entering it and one leaving it is called join. All flows going into the join of concurrent and parallel before processing may continue. This denotes the end of parallel processing.



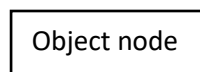
- **Decision:** The decision symbol is a diamond shape. It represents the branching or merging of various flows with the symbol acting as a frame or container.



- **Merge:** A diamond with several flows entering and one leaving. The representation is that one or more incoming flows must reach this point until processing continues.



- **Object node:** It is used to represent an object that is connected to a set of object flows



- **Activity diagram**

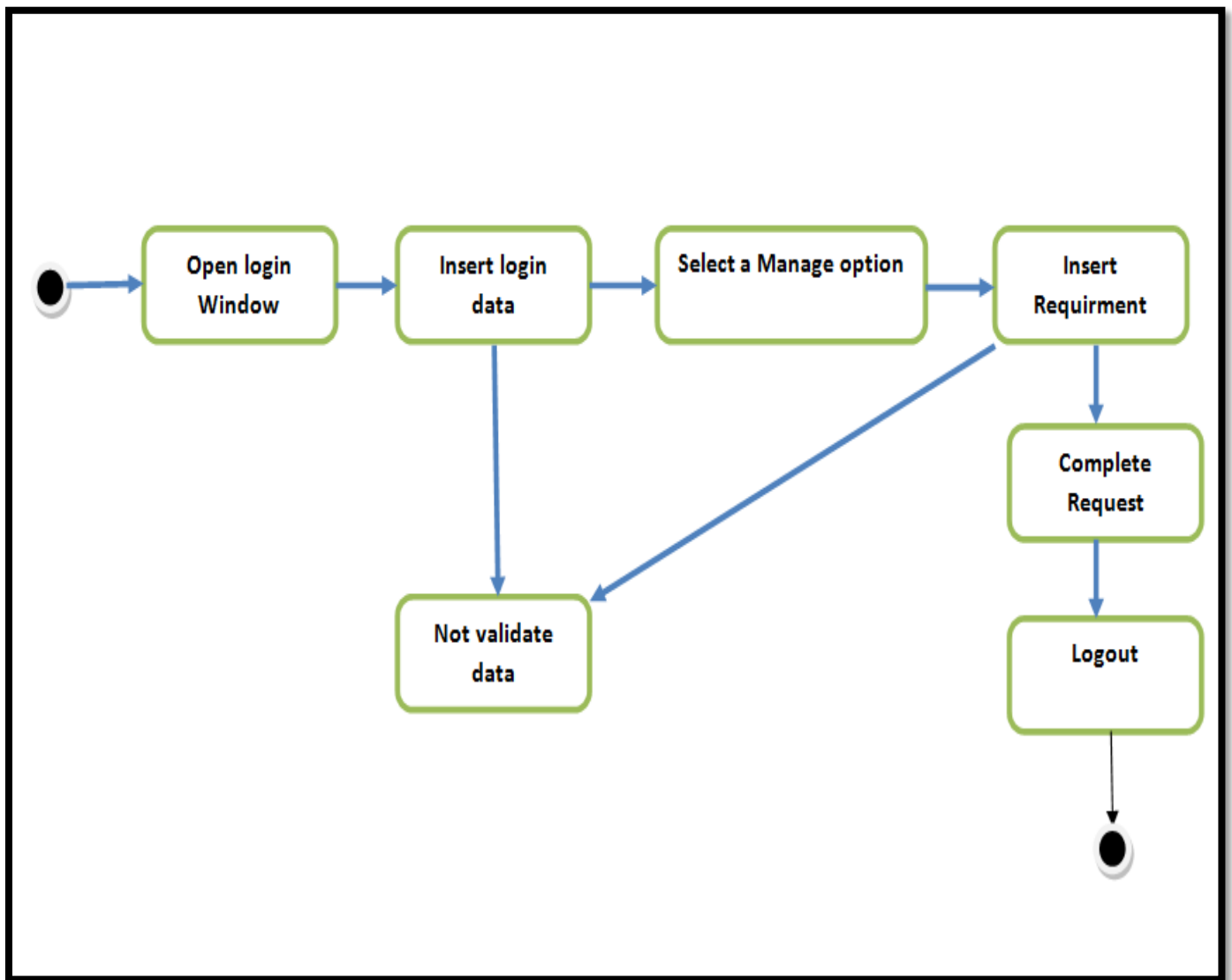
- Activity Diagrams describe the workflow behavior of the current system
- Activity diagrams are used in process data modeling and analysis using basic requirement engineering.
- A typical business process which analyze several external incoming events that can be represented by activity diagrams.

They are most useful for understanding work flow analysis of synchronous behaviors across a process.

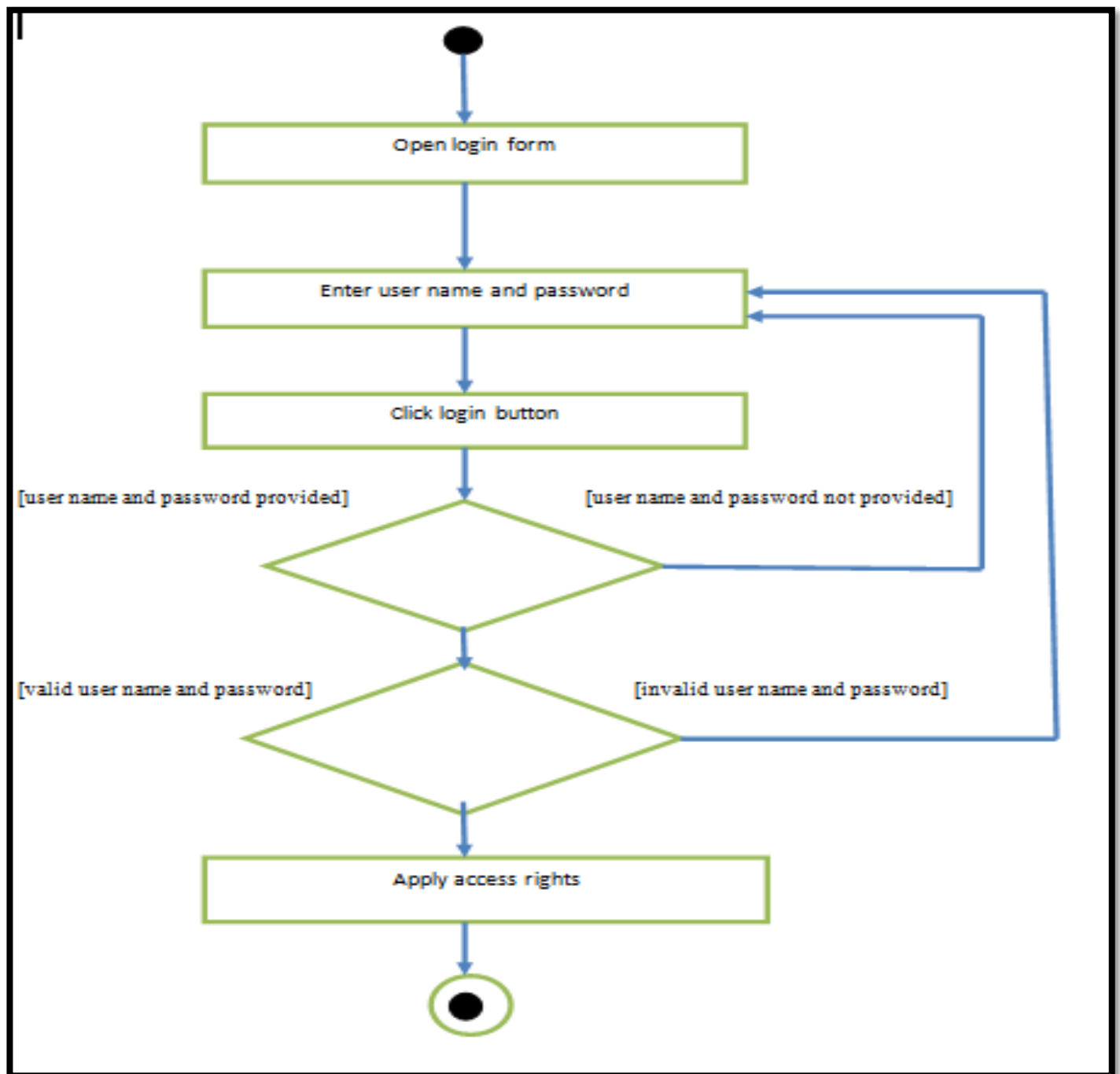
Activity diagrams are used for

- Documenting existing process
- Analyzing new process Concepts
- Finding reengineering opportunities

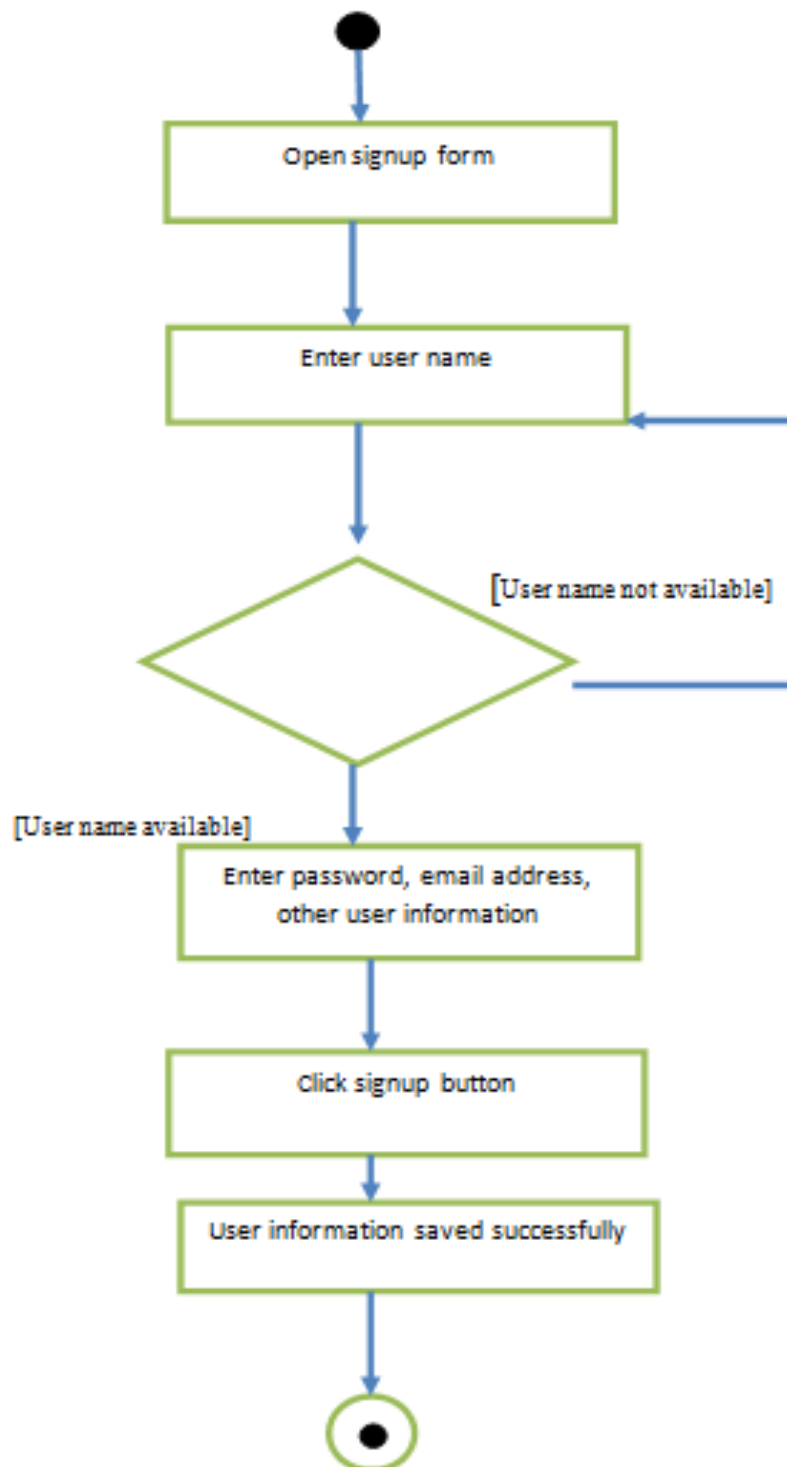
4.3.1 Activity Diagram:



4.3.2 Activity Diagram for Login



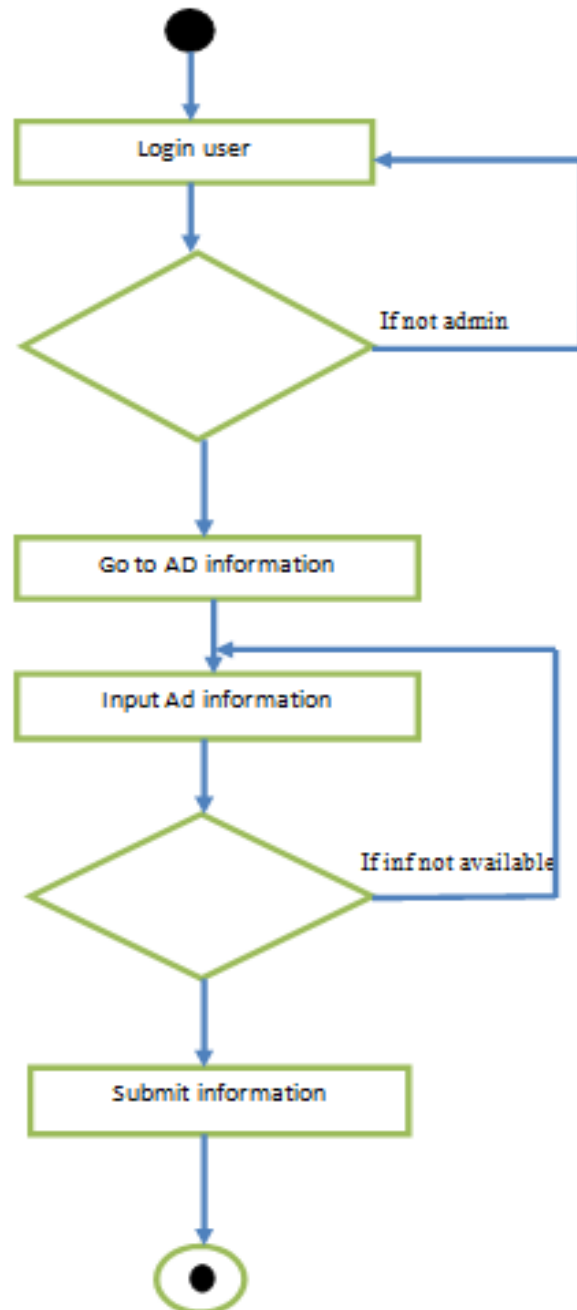
Signup:



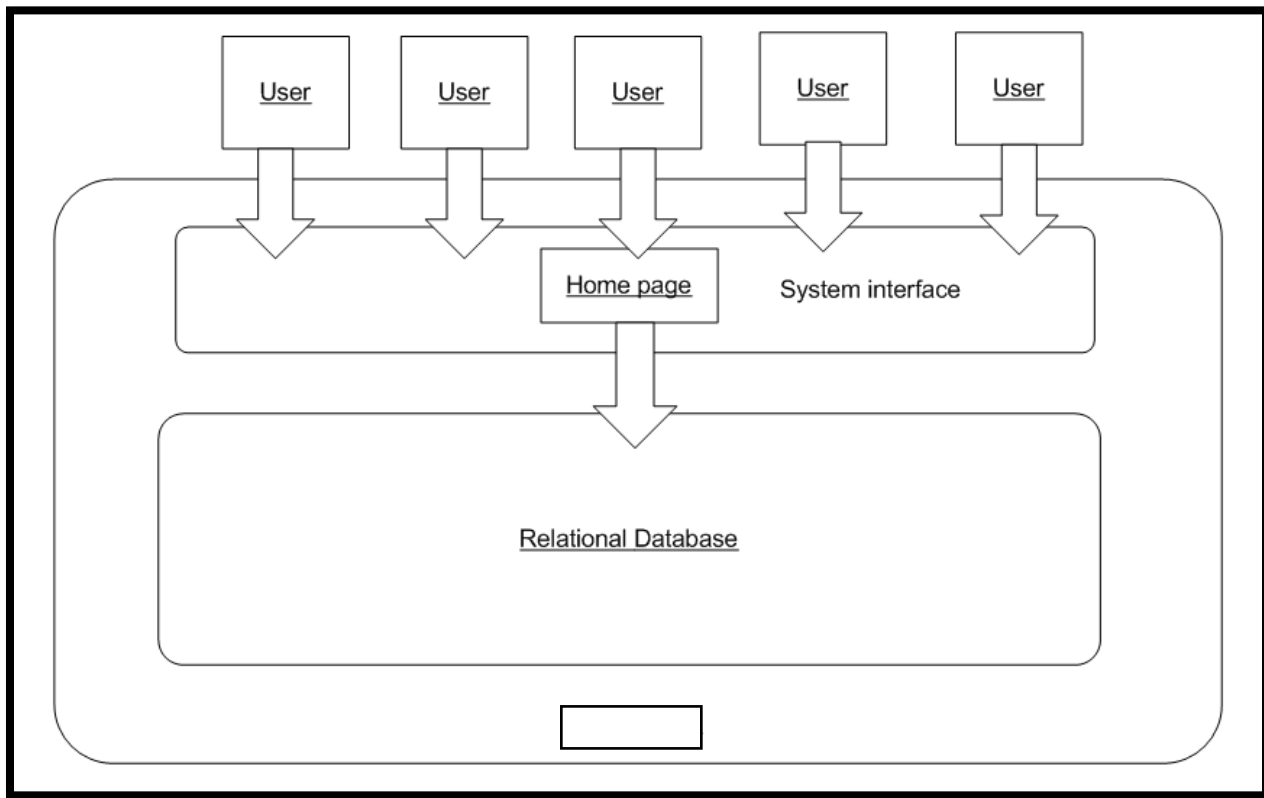
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4.3.3 Admin AD

Adding Ad:

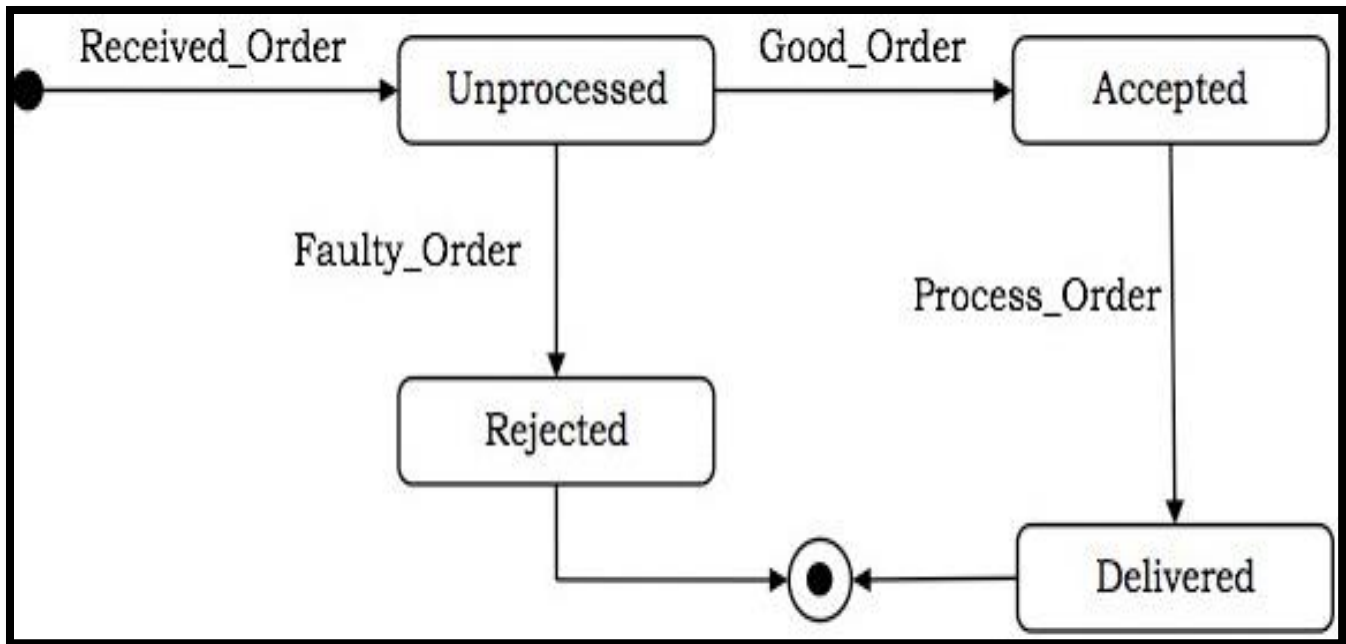


4.4 Block Diagram



4.5 State Transition Diagram

- The state transition diagram basically map out the state of the various system to show the essential changes in the framework and show the working of the framework.
- State transition diagrams have been used right from the beginning in object oriented modeling.
- The basic idea is to define a machine that has a number of states. This type of diagram essentially show the behavior of the system.
- The initial state is automatically created when the model is first created.
- An initial state is represented by a solid circular shaped node.
- There can be only one initial state of the system and it can't be deleted.
- An initial state may have no incoming transitions.



Starting Point	This is the place which speaks to the beginning phase of the framework. There might be a line with which advisers for the beginning stage.
States	States are appeared with the utilization of rectangles with the adjusted edges. These are part into two sections. Half has the name of the state. What's more, the other half contains the further qualities of the state.
Transitions	The moves are the bolts which are utilized to demonstrate the development of the state starting with one state then onto the next state in the framework.
Objects	These are the classes of the framework. They can contain many states and moves. These are appeared with the utilization of greater boxes.

